GOVERNMENT OF UTTARAKHAND • USDMA & ITDA DEVBHOOMI

AI Mountain Disaster Early-Warning Intelligence

UKIS 2026 Problem Statement P-011 • System Documentation & Live Links

EXECUTIVE SUMMARY (WHY THIS IS NOT JUST A DASHBOARD):
Mountain flash floods travel at >5 m/s, leaving only 15–45 minutes of warning time. This system actively defends lives by: **(1) Autonomously dialing voice phone calls** to SDRF & District Magistrates with spoken English/Hindi audio briefings, **(2) Pushing multilingual SMS** in 6 dialects with Dijkstra escape routes, **(3) Actuating 120dB physical sirens**, and **(4) Operating an interactive AI Commander Bot** with 0% data fabrication.

1. Live Clickable Links for Judges & Evaluators

Click any blue link below to open the live system directly in your browser:

Platform Feature	Clickable Live URL (Click to Open)	Live Interactive Capability
Citizen Active Alerts	https://frontend-rosy-phi-23.vercel.app	Official CAP v1.2 real-time emergency warnings (Chamoli, Rudraprayag, Uttarkashi)
Mountain Intelligence Hub	https://frontend-rosy-phi-23.vercel.app/mountain-intelligence	Live empirical modeling of GLOFs, cloudbursts, canyon hydraulics, seismic cascades & sensor mortality
Zero-Data Alert Simulator	https://frontend-rosy-phi-23.vercel.app/api/v1/alerts/mobile-broadcast	3GPP TS 23.041 zero-data & power-cut mobile broadcast test API (iOS & Android)
Interactive AI Commander Bot	https://frontend-rosy-phi-23.vercel.app	Click bottom-right floating bot. Answers evacuation queries, hazard reasons, and SOPs
ISRO Bhuvan GIS Map	https://frontend-rosy-phi-23.vercel.app/emergency	Interactive 14-hazard spatial grid, satellite/terrain layers & telemetry badges
Admin Room & SMS Dispatcher	https://frontend-rosy-phi-23.vercel.app/admin	Tab 7: Test live multilingual SMS push & dispatch automated authority voice calls
Evacuation Shelters Directory	https://frontend-rosy-phi-23.vercel.app/shelters	High-ground shelter capacities, GPS coordinates & emergency supply tracking
Citizen Guidebook	https://frontend-rosy-phi-23.vercel.app/guidebook	Offline survival protocols, flash flood evacuation rules & hotlines (112 / 1070)
Permanent Cloud Link (Vercel)	https://frontend-rosy-phi-23.vercel.app	Permanent production deployment hosted on Vercel's global edge network
GitHub Repository (Dual-Sync)	github.com/.../ukis-mountain-disaster-early-warning	Public repository with dual-sync script for automated Vercel production deployment

2. What Data Are We Curating & From Where?

Source Provider	Curated Parameters	Cadence	Core Function in Platform
Open-Meteo Weather API	Rainfall rate, 24h rain, 0°C freezing level, soil moisture	Live Real-Time	Rainfall rate-of-rise & cloudburst detection
Open-Meteo DEM Elevation	Copernicus / SRTM 10m Digital Elevation Model	Live DEM Query	Slope gradient, contour elevation & flood routing
Copernicus GloFAS (ECMWF)	River discharge forecasts (m³/s), 5-day runoff	Live Forecast	Alaknanda & Bhagirathi basin flood surge modeling
USGS Real-Time Seismicity	Earthquake magnitude, depth, epicenter coordinates	Live Feed	Slope creep stress & seismically-triggered landslide alerts
Central Water Commission	Gauge river stage (m), danger mark exceedance	Sensor Stream	River cresting detection & automated call triggers
USDMA Shelter Registry	Shelter capacities, high-ground GPS coordinates	GIS Registry	Dynamic haven allocation & shortest escape routing

3. Active Multi-Channel Alerting & Calling System

• **Autonomous Authority Voice Calls:** Dials emergency commanders (SEOC, SDRF, District Magistrates) with synthesized bilingual English/Hindi audio briefings and gathers keypad acknowledgment.

• **Hyperlocal Multilingual SMS Push:** Dispatches vernacular SMS in 6 languages (Hindi, Garhwali, Kumaoni, Jaunsari, English, Sanskrit) embedding dynamically computed safe haven escape routes.

• **C-DOT Cell Broadcast (Zero Data):** Mass geo-fenced cell tower pings alerting all mobile handsets in the hazard zone without requiring internet or active SIM balance.

• **Physical 120dB Sirens:** Actuates on-site relay horns and strobe beacons on sensor towers to wake sleeping residents during nocturnal mountain floods.

• **Interactive AI Commander Bot:** Answers operational queries strictly grounded in live telemetry and USDMA SOPs with 0% data fabrication.